

Hierarchical Shell-Manifold Theory

Geometric Foundations, Hierarchical Coupling, Projection, and Projected Effective Structure

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Abstract

Hierarchical Shell-Manifold Theory (HSMT) is a proposed geometric framework in which physical structure is organized by an integer-graded hierarchy of effective manifolds rather than by a single undifferentiated spacetime description. The central object is a graded space

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N,$$

whose neighboring grades are coupled through a localized overlap kernel. The framework introduces a continuous hierarchical coordinate as an organizing device, a Gaussian residual coupling between grades, a working graded operator, and a projection procedure by which an effective sector may be isolated from the larger hierarchy.

The present paper gives a self-contained formulation of HSMT at its current stage of development. The purpose is not to claim a completed fundamental physical theory, nor to derive the Standard Model, gravity, or cosmology from first principles. Instead, the paper establishes the architecture of HSMT, constructs an associative reference realization in which the principal mechanisms can be analyzed mathematically, and distinguishes clearly between definitions, reference-model results, working hypotheses, phenomenological targets, and unresolved derivations.

A central component of the framework is the projection of a distinguished grade, presently represented by $N = 4$, from the full graded system. The exact Feshbach–Nakajima–Zwanzig structure shows that the effective dynamics of the projected sector contain contributions from the eliminated grades through an energy-dependent self-energy or, equivalently, a time-domain memory kernel. In an appropriate isolation regime, the projected sector admits an effective Weyl-type description. A scale-dependent spatial symbol provides a mathematically explicit route to dimensional flow, while the distinction between effective temporal evolution and a thermodynamic arrow of time is maintained.

The present HSMT framework therefore constitutes a mathematical and conceptual foundation from which a physical theory may be developed. The principal open problems are the dynamical selection of the observed grade, the derivation of presently free scales and parameters,

the derivation of internal gauge and matter structure, the derivation of irreversible time, and the construction of a controlled continuum and gravitational limit.

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1 Introduction

Hierarchical Shell-Manifold Theory (HSMT) proposes that the structure ordinarily described as physical spacetime may be understood as a projected effective description of a deeper graded geometry. Rather than assuming from the outset that the physical world is represented by a single four-dimensional manifold with fixed dimensional and dynamical properties, HSMT introduces a hierarchy of related sectors,

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N, \quad (1)$$

where the integer N labels hierarchical grade.

The purpose of the hierarchical construction is not merely to introduce an additional discrete coordinate. The central hypothesis is that the grades are dynamically coupled, while an observer or effective low-energy description may access only a restricted sector of the full structure. Physical dynamics may therefore arise through the interaction of three levels:

fundamental graded geometry $\longrightarrow$ hierarchical coupling $\longrightarrow$ projected effective physics.

 (2)

This provides the organizing principle of HSMT.

The present paper is intended to establish HSMT as a coherent framework in its current state rather than to extend the theory prematurely into a complete unification program. Earlier HSMT work considered possible consequences for quantum mechanics, gauge structure, particle physics, gravity, cosmology, and other phenomena. Those possibilities remain part of the broader research program, but the present paper restricts itself to the structures that can currently be formulated directly and analyzed systematically.

The central mathematical question is therefore:

Can a self-adjoint graded operator with localized inter-grade coupling produce, after controlled projection onto an effective sector, the local dynamical structures observed in ordinary physics?

This question is narrower than the claim that HSMT has already reproduced known physics, but it is sufficiently precise to support mathematical investigation.

The framework developed below has several principal ingredients:

- (i) a discrete graded manifold;
- (ii) a continuous hierarchical coordinate used to organize the grade structure;
- (iii) localized Gaussian coupling between grades;
- (iv) a working graded operator, with a formal Dirac-type organizer retained only as a parent expression;

- (v) an associative reference realization suitable for rigorous operator analysis;
- (vi) projection onto a distinguished grade;
- (vii) exact effective dynamics obtained by eliminating the complementary grades;
- (viii) an effective Weyl-type sector in an isolation regime;
- (ix) scale-dependent spectral order and dimensional flow;
- (x) a distinction between effective time evolution and thermodynamic irreversibility.

The framework deliberately distinguishes between what has been defined, what follows mathematically within the reference model, what remains a working hypothesis, and what remains an open physical derivation.

2 Status and Scope of the Framework

HSMT is presently a framework for constructing a physical theory rather than a completed physical theory. This distinction is fundamental.

The current status can be summarized as

$$\boxed{\text{HSMT framework} \longrightarrow \text{reference realization} \longrightarrow \text{physical theory} \longrightarrow \text{empirical validation.}} \quad (3)$$

The first two stages are the present subject of this paper.

The transition to a complete physical theory requires additional derivations. In particular, presently free structures must ultimately be determined from the HSMT core rather than inserted independently. These include the physical value of the Gaussian width, the origin of the momentum interpolation scale, the dynamical selection of the observed grade, the origin of gauge structure, the matter content, the gravitational sector, and the mechanism responsible for irreversible macroscopic time.

Accordingly, the terminology used throughout this paper is:

- **Definition:** a structure explicitly specified as part of HSMT.
- **Reference-model result:** a mathematical result obtained within an explicitly defined associative realization.
- **Working hypothesis:** a proposed HSMT mechanism whose complete derivation remains unfinished.
- **Phenomenological target:** a known physical structure that HSMT seeks eventually to derive or reproduce.

- **Open derivation:** a result required for completion of the physical theory but not established here.

This classification is not merely editorial. It is part of the scientific structure of the theory.

3 Fundamental HSMT Geometry

3.1 The graded manifold

The fundamental geometric postulate is that the total structure is decomposed into integer-labeled grades:

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N. \quad (4)$$

Each $\mathcal{M}_N$ represents a grade of the hierarchy. The symbol N is not initially interpreted as an ordinary spatial coordinate. Rather, it labels distinct sectors of the total geometric construction.

The grades may possess related local spatial structures, but the total geometry is not assumed to be equivalent to an ordinary Cartesian product

$$\mathcal{M}_4 \times \mathbb{Z}.$$

The direct-sum or disjoint-union notation is therefore used to emphasize that the hierarchy is a structural organization of sectors.

3.2 Continuous hierarchical coordinate

A continuous coordinate ρ may be introduced to organize the hierarchy:

$$\rho \in \mathbb{R}. \quad (5)$$

The integer grades are then represented as localized regions or centers in the ρ direction. A basic localization profile is

$$w(\rho) = \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(-\frac{\rho^2}{2\sigma_0^2}\right), \quad (6)$$

where σ_0 is the characteristic width.

The present framework has considered

$$\sigma_0 = \frac{\sqrt{2}}{4} \quad (7)$$

as a candidate value.

Equation (7) is not currently derived from the fundamental theory and is therefore a candidate or test value rather than a prediction.

A convenient correspondence between grade and hierarchical coordinate is

$$\rho_N = N - 4. \quad (8)$$

The shift by four is presently a modeling convention associated with the distinguished grade $N = 4$. A future dynamical theory must determine whether this distinguished value follows from the structure or must be regarded as an external choice.

4 Gaussian Hierarchical Coupling

4.1 Discrete overlap kernel

The continuous Gaussian localization induces a natural discrete overlap structure. A basic unnormalized kernel is

$$W_{NM} = \exp \left[-\frac{(N - M)^2}{2\sigma_0^2} \right]. \quad (9)$$

A normalized discrete kernel may be defined through

$$Z_\sigma = \sum_{k \in \mathbb{Z}} \exp \left(-\frac{k^2}{2\sigma_0^2} \right), \quad (10)$$

with

$$p_k = \frac{\exp[-k^2/(2\sigma_0^2)]}{Z_\sigma}. \quad (11)$$

The corresponding convolution operator on $\ell^2(\mathbb{Z})$ is

$$(G_\sigma f)_N = \sum_{M \in \mathbb{Z}} p_{N-M} f_M. \quad (12)$$

Because p_k is real, symmetric, and belongs to $\ell^1(\mathbb{Z})$, G_σ is a bounded self-adjoint operator on $\ell^2(\mathbb{Z})$.

4.2 Nearest-neighbor approximation

The full Gaussian kernel is nonzero for every grade separation. Thus, strictly speaking, Gaussian coupling is not nearest-neighbor coupling.

A nearest-neighbor model can instead be defined as a controlled truncation,

$$G_{\text{NN}} = c_0 \mathbf{1} + c_1 (S + S^\dagger), \quad (13)$$

where

$$(Sf)_N = f_{N+1}. \quad (14)$$

For sufficiently localized Gaussian width, the $|N - M| = 1$ terms dominate over more distant

couplings. For the candidate value

$$\sigma_0 = \frac{\sqrt{2}}{4},$$

the unnormalized first-neighbor weight is

$$\exp\left(-\frac{1}{2\sigma_0^2}\right) = \exp(-4), \quad (15)$$

while the second-neighbor weight is

$$\exp\left(-\frac{4}{2\sigma_0^2}\right) = \exp(-16). \quad (16)$$

The nearest-neighbor approximation is therefore useful as a reference model, but the distinction between the exact Gaussian hierarchy and its truncation must be maintained.

4.3 Fourier representation in grade space

Let

$$f_N = \int_{-\pi}^{\pi} \frac{dq}{2\pi} \tilde{f}(q) e^{iqN}. \quad (17)$$

Then the normalized Gaussian convolution has multiplier

$$g_\sigma(q) = \sum_{k \in \mathbb{Z}} p_k e^{iqk}. \quad (18)$$

Because $p_k = p_{-k}$,

$$g_\sigma(q) = \sum_{k \in \mathbb{Z}} p_k \cos(kq). \quad (19)$$

For long wavelengths,

$$g_\sigma(q) = 1 - \frac{\mu_2}{2} q^2 + \frac{\mu_4}{24} q^4 + \mathcal{O}(q^6), \quad (20)$$

where

$$\mu_{2n} = \sum_{k \in \mathbb{Z}} p_k k^{2n}. \quad (21)$$

Equation (20) gives a concrete route toward a continuum description of the hierarchy. Locality in the effective hierarchical direction is expected only in an appropriate long-wavelength regime.

5 The Master Operator

5.1 Working graded organizer

A formal Dirac-type operator can be written in the hierarchical coordinate as

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \frac{A}{2} (L_u + R_u) \sigma^2 + m_0 \sigma^3, \quad (22)$$

where L_u and R_u denote left and right multiplication by an internal element u , and σ^i are Pauli matrices.

Equation (22) serves as a formal organizer for the hierarchical construction. It should not be interpreted as a completed fundamental operator until its domain, internal algebra, spectrum, and relation to the associative reference realization have been fully specified.

The main mathematical laboratory used in the present framework is instead an associative Hilbert-space realization.

6 Associative Reference Realization

6.1 Hilbert space

The reference Hilbert space used in this paper is the product laboratory

$$\mathcal{H} = L^2(\mathbb{R}^3, \mathbb{C}^2) \otimes \ell^2(\mathbb{Z}). \quad (23)$$

An equivalent shared-fibre laboratory $L^2(\mathbb{R}^3 \times \mathbb{Z}, \mathbb{C}^2)$ may be used when spatial and hierarchical operators are required to act on one spinor. The two realizations are calculational gauges, not competing ontologies. The product form is kept here because the Kato–Rellich argument is then immediate.

A state may be written as

$$\Psi = \{\psi_N(\mathbf{x})\}_{N \in \mathbb{Z}}, \quad (24)$$

with

$$\psi_N \in L^2(\mathbb{R}^3, \mathbb{C}^2). \quad (25)$$

The spatial operator is initially taken to be the Weyl operator

$$D_{\text{sp}} = -i\boldsymbol{\sigma} \cdot \nabla. \quad (26)$$

On the standard Sobolev domain,

$$D_{\text{sp}} : H^1(\mathbb{R}^3, \mathbb{C}^2) \rightarrow L^2(\mathbb{R}^3, \mathbb{C}^2),$$

the operator is self-adjoint.

6.2 Reference Hamiltonian

A representative reference Hamiltonian is

$$H = D_{\text{sp}} + H_{\text{grade}} + H_{\text{hop}} + H_{\text{int}}. \quad (27)$$

The grade-dependent on-site term may be chosen as

$$H_{\text{grade}} = \sum_{N \in \mathbb{Z}} W_N \Phi_N, \quad (28)$$

where Φ_N projects onto grade N .

An explicit isolation profile is

$$W_N = \kappa \tanh[\alpha(N - 4)]. \quad (29)$$

This profile is odd about $N = 4$:

$$W_{4+r} = -W_{4-r}. \quad (30)$$

The parameters κ and α determine the scale and sharpness of the grade separation.

The hierarchical hopping term may be written

$$H_{\text{hop}} = t \sum_N (\Phi_{N+1,N} + \Phi_{N,N+1}), \quad (31)$$

or more generally by the Gaussian kernel (9).

A useful parameterization is

$$t = t_0 \exp\left(-\frac{1}{4\sigma_0^2}\right), \quad (32)$$

but this relation is a model convention unless derived from a specified overlap integral.

6.3 Self-adjointness

If D_{sp} is self-adjoint and the remaining grade and hopping terms are bounded self-adjoint operators, then the full operator is self-adjoint on the domain of D_{sp} by the Kato–Rellich theorem.

This provides an important mathematical baseline:

$$\boxed{\text{well-defined reference dynamics} \implies \text{self-adjoint reference operator.}} \quad (33)$$

For scale-dependent operators of order $\gamma > 1$, the appropriate spectral domain must instead be defined through functional calculus. Such operators generally require higher regularity than H^1 .

7 Distinguished Grade and the Meaning of $N = 4$

The present HSMT construction distinguishes the grade

$$N = 4. \quad (34)$$

The reason is that the reference model is constructed to investigate whether a four-dimensional effective physical sector can be isolated from a broader hierarchy.

It is important to state the current status precisely:

$$N = 4 \text{ is presently a distinguished modeling choice, not a completed dynamical selection theorem.} \quad (35)$$

The theory must eventually determine whether the observed grade follows from the dynamics.

A future selection mechanism could seek a grade $N_\star$ satisfying several simultaneous requirements:

1. a stable spectral isolation gap;
2. a controlled projection error;
3. an acceptable infrared spatial dimension;
4. stable effective propagation;
5. a suitable continuum limit;
6. retention of sufficient information from the surrounding hierarchy.

Rather than imposing an arbitrary numerical score, HSMT should seek a mathematically defined selection criterion derived from the operator itself.

The desired future result is therefore not merely

$$N = 4,$$

but

$$N_\star = 4 \quad (36)$$

as the solution of an independently defined spectral or variational problem.

8 Projection and Effective Dynamics

8.1 Projection operators

Let

$$P = \Phi_4, \quad Q = \mathbf{1} - P. \quad (37)$$

Any state may be decomposed as

$$|\Psi\rangle = P|\Psi\rangle + Q|\Psi\rangle. \quad (38)$$

Writing

$$|\Psi\rangle = |\psi_4\rangle + |\chi_\perp\rangle, \quad (39)$$

makes explicit that the observed sector is embedded in a larger state space.

8.2 Exact projected resolvent

With respect to the decomposition

$$\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H},$$

the Hamiltonian has block form

$$H = \begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix}. \quad (40)$$

The exact projected resolvent is

$$\boxed{P(z - H)^{-1}P = \left[z - H_{PP} - H_{PQ}(z - H_{QQ})^{-1}H_{QP} \right]^{-1}.} \quad (41)$$

The second term,

$$\Sigma(z) = H_{PQ}(z - H_{QQ})^{-1}H_{QP}, \quad (42)$$

is the energy-dependent self-energy generated by the eliminated grades.

This is a central structural result of the reference realization.

The effective sector is therefore not generally governed by H_{PP} alone.

8.3 Exact time-domain structure

Assume for simplicity that

$$Q\Psi(0) = 0.$$

Eliminating the Q sector gives an exact equation of the form

$$i \frac{d}{dt} P\Psi(t) = H_{PP}P\Psi(t) - i \int_0^t d\tau H_{PQ}e^{-iH_{QQ}\tau}H_{QP}P\Psi(t - \tau). \quad (43)$$

The second term is a memory contribution.

Thus projection produces, in general,

$$\boxed{\text{effective dynamics} = \text{direct dynamics} + \text{memory from eliminated grades.}} \quad (44)$$

A local-in-time effective Hamiltonian appears only under additional approximations such as a suitable isolation limit or controlled Markov approximation.

9 Isolation of the Effective Sector

Suppose an energy interval I of interest is separated from the spectrum of H_{QQ} by

$$\Delta = \text{dist}(I, \text{spec}(H_{QQ})) > 0. \quad (45)$$

Then, for z sufficiently close to the interval but outside $\text{spec}(H_{QQ})$,

$$\|\Sigma(z)\| \leq \frac{\|H_{PQ}\|^2}{\text{dist}(z, \text{spec}(H_{QQ}))}. \quad (46)$$

This gives a natural isolation parameter:

$$\varepsilon_{\text{iso}} = \frac{\|H_{PQ}\|}{\Delta}. \quad (47)$$

The effective-sector approximation is controlled when

$$\varepsilon_{\text{iso}} \ll 1. \quad (48)$$

This establishes a concrete mathematical meaning for the phrase “isolation of the $N = 4$ shell.”

It is not assumed that the surrounding grades disappear. Instead, their influence becomes perturbatively controlled.

10 Three-Grade Reference Model

For analytical purposes, consider the local three-grade system

$$N = 3, 4, 5.$$

Let the central grade have zero on-site offset and the neighbors have opposite offsets:

$$W_4 = 0, \quad W_3 = -v, \quad W_5 = +v. \quad (49)$$

A schematic momentum-space matrix, with internal spin structure suppressed, is

$$H(\mathbf{k}) = \begin{pmatrix} -|k| + W_3 & t & 0 \\ t & |k| & t \\ 0 & t & |k| + W_5 \end{pmatrix}. \quad (50)$$

This matrix is a local three-grade sketch. It is not the spectrum of the full reference operator.

Elimination of the two neighboring sectors generates a central-sector self-energy.

In a symmetric approximation, a representative frequency-dependent structure takes the form

$$\Sigma(k, \omega) = \frac{2\omega t^2}{\omega^2 - |k|^2 - v^2}. \quad (51)$$

For

$$|\omega|, |k| \ll v,$$

one obtains

$$\Sigma(k, \omega) = -\frac{2\omega t^2}{v^2} + \mathcal{O}\left(\frac{\omega(k^2 + \omega^2)}{v^4}\right). \quad (52)$$

Thus the principal low-energy effect of the neighboring grades is a renormalization of the central temporal coefficient together with higher-order corrections.

The exact coefficient depends on the detailed convention for the three-grade Hamiltonian. The structural conclusion is more important:

finite grade coupling + large separation $\rightarrow$ controlled effective central-sector dynamics.

(53)

11 Effective Weyl Sector

The reference model allows a two-component central-sector description after projection and isolation.

For an isolated grade, the effective spatial operator may be represented as

$$H_4 = Z|\mathbf{p}|^{\gamma(|\mathbf{p}|)-1} \boldsymbol{\sigma} \cdot \mathbf{p}, \quad (54)$$

where

$$Z = 1 - \frac{t^2}{v^2} \quad (55)$$

in a representative symmetric isolation approximation.

The corresponding effective equation is

$$i\partial_t \psi = H_4 \psi. \quad (56)$$

Equation (56) should be interpreted as an effective projected evolution equation. It is not being asserted as the fundamental equation of HSMT.

In the infrared limit,

$$\gamma(|\mathbf{p}|) \rightarrow 1, \quad (57)$$

so that

$$H_4 \rightarrow Z \boldsymbol{\sigma} \cdot \mathbf{p}. \quad (58)$$

The multiplicative factor Z can be absorbed into an effective temporal normalization or velocity convention.

Thus ordinary Weyl propagation appears as the low-energy limit of the isolated reference sector.

12 Scale-Dependent Spectral Order

12.1 Homogeneous symbols

Consider a spatial operator with dispersion

$$E(\mathbf{k}) \sim |\mathbf{k}|^\gamma. \quad (59)$$

The corresponding spatial heat kernel has the scaling

$$K_s(\tau) \propto \int_{\mathbb{R}^3} d^3k \exp\left(-\tau|\mathbf{k}|^{2\gamma}\right). \quad (60)$$

Under the rescaling

$$\mathbf{k} = \tau^{-1/(2\gamma)} \mathbf{q},$$

one obtains

$$K_s(\tau) \propto \tau^{-3/(2\gamma)}. \quad (61)$$

The spectral dimension is defined by

$$d_s = -2 \frac{d \ln K_s}{d \ln \tau}. \quad (62)$$

Therefore

$$\boxed{d_s = \frac{3}{\gamma}}. \quad (63)$$

This is a spatial spectral dimension for the homogeneous spatial operator.

12.2 Running order

A scale-dependent order may be introduced through

$$\gamma(k) = 1 + (\gamma_{\text{UV}} - 1) \frac{(k/k_*)^2}{1 + (k/k_*)^2}. \quad (64)$$

The infrared limit is

$$\gamma(k) \rightarrow 1, \quad k \ll k_*, \quad (65)$$

while

$$\gamma(k) \rightarrow \gamma_{\text{UV}}, \quad k \gg k_*. \quad (66)$$

If

$$\gamma_{\text{UV}} = \frac{3}{2}, \quad (67)$$

then the homogeneous scaling relation gives

$$d_s^{\text{IR}} = 3, \quad d_s^{\text{UV}} = 2. \quad (68)$$

These are spatial spectral dimensions.

No claim is made here that the full spacetime spectral dimension is automatically two or three. Such a conclusion requires the temporal sector to be specified and analyzed consistently.

12.3 Running heat kernel

For a genuinely scale-dependent $\gamma(k)$, the exact simple power law no longer applies. One must instead evaluate

$$K_s(\tau) \propto \int_{\mathbb{R}^3} d^3k \exp \left[-\tau |\mathbf{k}|^{2\gamma(|\mathbf{k}|)} \right], \quad (69)$$

and define

$$d_s(\tau) = -2 \frac{d \ln K_s(\tau)}{d \ln \tau}. \quad (70)$$

This provides a direct computational test of the dimensional-flow hypothesis.

13 The Interpolation Scale

The scale k_* determines the momentum range over which the effective spatial order changes.

At the present stage,

$$k_* \quad (71)$$

is an input parameter.

It is not derived from σ_0 , the grade gap v , or the hopping t .

In particular, the following identifications are not presently established:

$$k_* = \frac{1}{\ell_P}, \quad k_* = v, \quad k_* = t, \quad (72)$$

nor is k_* currently fixed by the condition

$$d_s^{\text{UV}} = 2.$$

A completed HSMT theory should seek an equation of the form

$$\mathcal{F}(k_*, \sigma_0, v, t, \dots) = 0, \quad (73)$$

derived from the underlying operator or spectral structure.

Possible mechanisms include:

1. a spectral gap equation;
2. a fixed point of an explicitly defined renormalization flow;
3. a dynamically generated spectral cutoff;
4. a geometric scale generated by the fundamental measure.

None is presently established.

14 Effective Time and the Arrow of Time

14.1 Effective time

The HSMT framework distinguishes the parameter used to describe projected evolution from a fundamental assumption that time is an independent geometric coordinate.

Once the projected sector admits an effective equation

$$i\partial_t\psi = H_{\text{eff}}\psi, \quad (74)$$

the parameter t labels effective non-stationary evolution of the projected state.

The conceptual proposal is therefore

$$\boxed{\text{effective time} \sim \text{parameterization of projected non-stationarity.}} \quad (75)$$

This should not be confused with the thermodynamic arrow of time.

14.2 Why a symmetric Gaussian is insufficient

The Gaussian kernel

$$W_{NM} = W_{MN}$$

is symmetric under grade reversal.

Consequently, the Gaussian overlap by itself does not establish a preferred temporal direction.

A temporal arrow requires additional structure, such as:

- inequivalent neighboring grades;
- asymmetric environmental spectral densities;
- coarse-graining;
- projection-induced information loss;
- or an appropriate reduced dynamical semigroup.

The odd profile

$$W_N = \kappa \tanh[\alpha(N - 4)] \quad (76)$$

provides a concrete way of making the neighboring grades dynamically inequivalent, but inequivalent neighbors alone do not constitute a complete derivation of the thermodynamic arrow.

14.3 Reduced open-system dynamics

Let the central grade be the system and neighboring grades the environment:

$$H = H_S + H_E + V. \quad (77)$$

With

$$H_S = \Phi_4 H \Phi_4, \quad (78)$$

and

$$H_E = (\Phi_3 + \Phi_5) H (\Phi_3 + \Phi_5), \quad (79)$$

the interaction is

$$V = V_3 + V_5. \quad (80)$$

Under the additional Born–Markov and secular assumptions, a reduced master equation may take the form

$$\frac{d\rho_4}{dt} = -i[H_S + H_{LS}, \rho_4] + \mathcal{D}_3[\rho_4] + \mathcal{D}_5[\rho_4]. \quad (81)$$

The corresponding spectral densities for homogeneous spatial slices of orders γ_α scale as

$$\nu_\alpha(\omega) \propto |\omega|^{3/\gamma_\alpha - 1}. \quad (82)$$

For example, if

$$\gamma_5 = 1, \quad \gamma_3 = \frac{3}{2},$$

then

$$\nu_5(\omega) \propto |\omega|^2, \quad \nu_3(\omega) \propto |\omega|. \quad (83)$$

This demonstrates how asymmetric neighboring sectors could influence reduced dynamics.

It does not yet constitute a complete HSMT derivation of the thermodynamic arrow. A rigorous entropy-production result requires the relevant bath assumptions, state class, finite-volume or trace-class conditions, and dynamical semigroup structure to be explicitly established.

15 The Geometric Meaning of Projection

HSMT uses “projected effective sector” in a specific mathematical sense. The word “emergence” is avoided except as a target of the later physical-theory stage.

The effective sector is not assumed to be fundamental. It is obtained by restricting the full state space through a projection

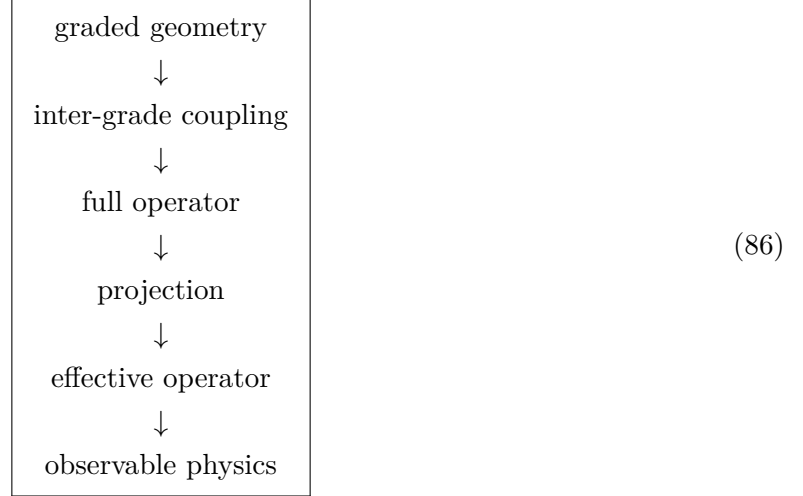
$$P : \mathcal{H} \rightarrow P\mathcal{H}. \quad (84)$$

The effective operator contains information about the complementary space through

$$H_{\text{eff}}(z) = H_{PP} + H_{PQ}(z - H_{QQ})^{-1}H_{QP}. \quad (85)$$

Therefore the effective theory can contain structures that are not explicit in the isolated central block.

The general HSMT projection chain is



This is the principal conceptual mechanism of the framework.

16 What Is Actually New in HSMT?

HSMT does not claim that every mathematical ingredient appearing in the framework is individually new. Gaussian kernels, graded Hilbert spaces, Dirac-type operators, spectral dimension, projection methods, effective Hamiltonians, and open-system dynamics all have established mathematical and physical precedents.

The proposed novelty lies instead in their organization into a single hierarchical framework.

The distinctive structural combination is:

integer-graded geometry + localized Gaussian coupling
 + projected effective sector
 + scale-dependent spectral order
 + effective temporal parameter.

(87)

In particular, HSMT proposes that these structures should not be treated as unrelated phenomenological modifications.

Instead, they form a dependency chain:

$$\text{hierarchy} \rightarrow \text{coupling} \rightarrow \text{projection} \rightarrow \text{effective dynamics} \rightarrow \text{scale dependence.} \quad (88)$$

The central scientific question is whether this chain can ultimately be closed without introducing independent structures at every stage.

17 Continuum and Locality

A central requirement for HSMT is recovery of ordinary local physics in an appropriate limit.

Let the hierarchical coordinate be written

$$\rho = Na_\rho, \quad (89)$$

where a_ρ is a hierarchical spacing.

The discrete Gaussian multiplier can be expanded at small hierarchical momentum q :

$$g_\sigma(q) = 1 - \frac{\mu_2}{2}q^2 + \mathcal{O}(q^4). \quad (90)$$

This suggests that a continuum description may be available when

$$|q|\sigma_0 \ll 1. \quad (91)$$

The continuum limit is therefore not automatic. It must satisfy at least the following conditions:

1. boundedness of the hierarchy operator;
2. convergence of the discrete spectrum to a controlled continuum spectrum;
3. suppression of nonlocal corrections at the scales of interest;
4. recovery of the desired effective spatial locality;
5. stability of the projected sector.

A complete continuum theorem remains an open problem.

18 Gauge Structure as a Downstream Extension

Gauge fields are not derived in the present framework. They are introduced as a later extension of the isolated sector.

For an Abelian connection A , define

$$\Pi = -i\nabla - eA, \quad (92)$$

and

$$T = \boldsymbol{\sigma} \cdot \Pi. \quad (93)$$

A natural functional-calculus extension of the effective Weyl operator is

$$H(T) = ZT|T|^{\gamma(|T|)-1}. \quad (94)$$

An alternative ordering would be

$$H(\Pi) = Z|\Pi|^{\gamma(|\Pi|)-1}T. \quad (95)$$

The two expressions agree in the free case but need not agree when gauge fields are nonzero because

$$T^2 = \Pi^2 + e \boldsymbol{\sigma} \cdot \mathbf{B} \quad (96)$$

in the Abelian case, with additional commutator structure for non-Abelian connections.

The choice between these calculi is therefore a genuine operator-ordering question.

The present framework does not select one uniquely.

Likewise, HSMT does not yet derive

$$SU(3) \times SU(2) \times U(1),$$

its coupling constants, or the full gauge field content.

These remain targets for the physical-theory stage.

19 Matter, Chirality, and Flavour

The same distinction applies to matter structure.

An intrinsic two-component Weyl sector arises naturally in the associative reference model. However, this does not by itself derive the full Standard Model fermion content.

A Dirac completion may be written schematically as

$$\Psi = \begin{pmatrix} \psi \\ \chi \end{pmatrix}, \quad (97)$$

with

$$H_D = \begin{pmatrix} H_4[\psi] & M \\ M^\dagger & -H_4[\chi] \end{pmatrix}. \quad (98)$$

Chiral projectors are

$$P_L = \frac{1 - \gamma^5}{2}, \quad P_R = \frac{1 + \gamma^5}{2}. \quad (99)$$

A Higgs-like field Φ can then be introduced through a Yukawa structure

$$\mathcal{L}_Y = -\bar{\psi}_L Y \Phi \psi_R + \text{h.c.} \quad (100)$$

After a vacuum expectation value,

$$\langle \Phi \rangle = \frac{v_H}{\sqrt{2}}, \quad (101)$$

the mass matrix becomes

$$M = \frac{v_H}{\sqrt{2}} Y. \quad (102)$$

At the present stage, Y is an independent matrix.

Likewise, introducing three copies

$$\Psi^{(i)}, \quad i = 1, 2, 3, \quad (103)$$

is a postulate rather than a derived result.

Therefore the present HSMT framework should not claim that three generations, Yukawa hierarchies, mixing angles, or the observed fermion masses have been derived.

The eventual physical theory must seek a geometric origin for these structures.

20 Gravity and Geometric Dynamics

Gravity represents one of the major future stages of HSMT development.

A possible route is through a spectral or induced-gravity construction in which an effective action takes a schematic form such as

$$\Gamma_{\text{eff}} \sim -\frac{1}{2} \log \det(\mathcal{D}^2). \quad (104)$$

A heat-kernel expansion could then generate geometric invariants of the effective metric.

However, a complete derivation requires a specified geometric algebra, operator, measure, and asymptotic expansion. The present HSMT framework does not yet provide a completed spectral triple or equivalent gravitational construction.

Accordingly,

gravity is a future derivational target, not a present result of HSMT.

(105)

The same principle applies to cosmological applications.

21 Non-Associative and Octonionic Extensions

HSMT has considered non-associative and octonionic structures as possible extensions.

Such structures are potentially relevant to the internal algebraic organization of the hierarchy, but they should not be used as substitutes for an explicit operator construction.

A non-associative extension must specify:

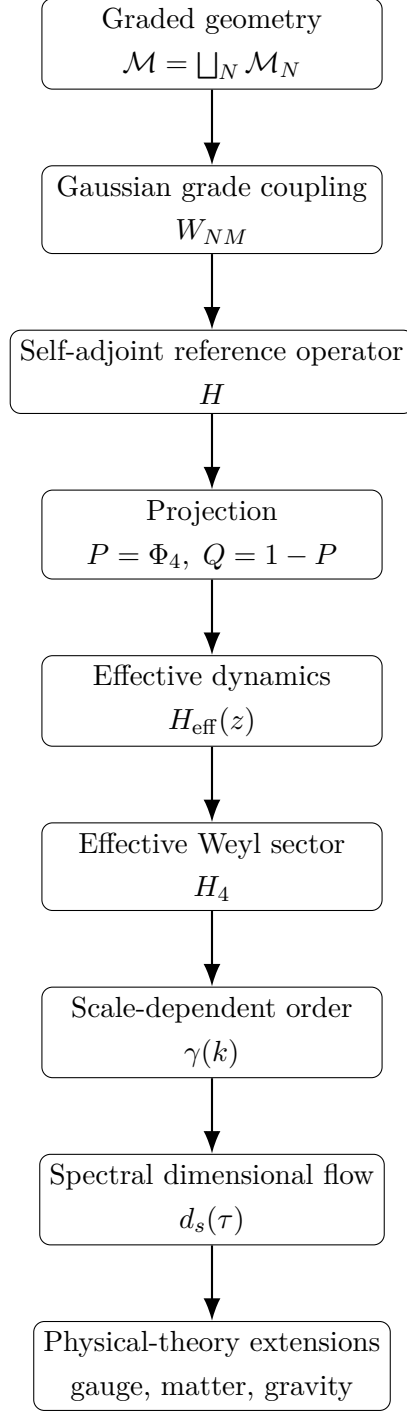
1. the state space;
2. the operator domain;
3. the multiplication rules;

4. the appropriate notion of adjointness;
5. the spectral problem;
6. the relation to the associative reference model.

Until those structures are established, the associative realization remains the principal mathematical laboratory.

22 A Unified Dependency Structure

The current HSMT architecture can be represented as follows.



The lower portion of this diagram contains structures that remain partly open.

The most important conceptual point is that the framework is intended to be hierarchical not only in its geometry but also in its derivations.

23 Claim and Derivation Ledger

A central methodological requirement of HSMT is that every substantive statement be assigned a clear status.

Structure	Status	Present meaning
$\mathcal{M} = \sqcup_N \mathcal{M}_N$	Definition	Fundamental graded organization of HSMT.
Gaussian overlap	Definition / ansatz	Localized coupling kernel between grades.
$\sigma_0 = \sqrt{2}/4$	Candidate	Test value; not presently derived.
Associative Hilbert space	Reference realization	Mathematical laboratory for explicit analysis.
Self-adjoint reference operator	Reference-model result	Available under stated domain and boundedness assumptions.
$N = 4$	Distinguished modeling choice	Candidate effective physical grade; dynamical selection remains open.
Feshbach projection	Derived formal result	Exact projected resolvent and effective self-energy.
Memory kernel	Derived formal result	Exact consequence of eliminating complementary grades.
Effective Weyl sector	Reference-model result	Obtained in the isolation regime.
$\gamma(k)$	Working ansatz	Scale-dependent spectral order.
k_*	Free input	Momentum scale of dimensional crossover.
$d_s = 3/\gamma$	Derived scaling result	Spatial spectral dimension for homogeneous spatial symbols.
$d_s^{UV} = 2$	Target	Obtained for $\gamma_{UV} = 3/2$.
Effective time	Working hypothesis	Projected non-stationarity interpreted as effective temporal evolution.
Thermodynamic arrow	Open derivation	Requires reduced irreversible dynamics and additional hypotheses.
$SU(3) \times SU(2) \times U(1)$	Extension	Gauge structure not presently derived.
Three generations	Extension	Matter multiplicity not presently derived.
Yukawa matrices	Extension	Mass and flavour matrices currently inserted phenomenologically.
Gravity	Future program	Requires a complete geometric dynamical construction.

This ledger should be treated as part of the framework rather than as a disclaimer appended to it.

24 Falsifiability and Failure Criteria

A useful theoretical framework must admit the possibility that its central construction fails.

HSMT therefore identifies several explicit failure modes.

24.1 Operator failure

If no mathematically well-defined self-adjoint reference operator can be constructed with the required hierarchical structure, the present formulation fails.

24.2 Isolation failure

If

$$\frac{\|H_{PQ}\|}{\Delta} \tag{106}$$

cannot be made small in the relevant physical regime, then the effective isolated-sector description is not controlled.

24.3 Continuum failure

If the discrete hierarchy does not possess an acceptable continuum or long-wavelength limit, HSMT cannot recover ordinary local physics in the proposed manner.

24.4 Dimensional-flow failure

If the running operator cannot generate a controlled spectral crossover while preserving the required infrared structure, the dimensional-flow mechanism fails.

24.5 $N = 4$ selection failure

If a complete dynamical selection problem is formulated and does not distinguish $N = 4$ under reasonable assumptions, then the present interpretation of the observed grade must be revised.

24.6 Temporal-interpretation failure

If projected non-stationarity cannot produce an effective temporal description compatible with the required dynamics, the proposed interpretation of time must be revised.

24.7 Gauge and matter failure

If gauge and matter structures cannot be derived consistently from a common HSMT extension, then those structures must remain external phenomenological additions rather than consequences of the theory.

These failure criteria make HSMT more testable as a mathematical research program.

25 What Would Constitute a Physical Theory?

The transition from framework to physical theory requires closure of the presently open structures.

A completed theory should ideally provide a chain of the form

$$\boxed{\{\mathcal{M}, \mathcal{D}, W\} \implies \{\text{parameters, dynamics, observed sector, interactions, predictions}\}.} \quad (107)$$

In particular, the following should ultimately be derived or constrained:

1. the physical value of σ_0 ;
2. the grade-selection mechanism;
3. the physical origin of k_* ;
4. the exact effective temporal structure;
5. the mechanism of irreversible time;
6. the internal gauge structure;
7. the matter representation content;
8. the origin of flavour and Yukawa parameters;
9. the gravitational dynamics;
10. the quantitative phenomenological predictions.

The theory should also specify the domain of validity of its effective descriptions and provide quantitative error estimates.

26 Research Program

The next stage of HSMT development can be organized into a sequence of mathematical problems.

26.1 Problem I: Complete the reference operator

Specify:

$$\text{Dom}(H), \quad \text{spec}(H), \quad \|H\|, \quad (108)$$

where appropriate, and establish self-adjointness and spectral properties for the full Gaussian hierarchy.

26.2 Problem II: Exact Gaussian hierarchy

Replace the nearest-neighbor truncation by the full kernel

$$W_{NM} = \exp \left[-\frac{(N-M)^2}{2\sigma_0^2} \right], \quad (109)$$

and quantify the difference between the exact and truncated systems.

26.3 Problem III: Controlled projection

Derive explicit bounds on

$$H_{\text{eff}}(z) \quad (110)$$

and establish the regime in which a local-in-time effective Hamiltonian is justified.

26.4 Problem IV: Dynamical grade selection

Construct a mathematically natural selection principle and determine whether it produces a unique distinguished grade.

The target result is

$$N_{\star} = 4, \quad (111)$$

but the result must follow from the theory rather than being imposed.

26.5 Problem V: Derive the dimensional flow

Replace the phenomenological $\gamma(k)$ by a function generated by the underlying operator, measure, renormalization flow, or spectral structure.

26.6 Problem VI: Derive effective time

Determine whether the effective temporal parameter can be obtained directly from the projected dynamics rather than introduced as an independent structure.

26.7 Problem VII: Derive irreversibility

Construct the appropriate reduced dynamics and determine whether entropy production or another invariant notion of irreversibility follows from the hierarchy.

26.8 Problem VIII: Internal structure

Investigate whether gauge groups, representations, chirality, and flavour can arise from the internal structure of the HSMT operator.

26.9 Problem IX: Gravitational sector

Construct a geometric action or equivalent dynamical principle and determine whether an Einstein-like low-energy limit emerges.

26.10 Problem X: Quantitative predictions

Only after the previous stages are established should numerical predictions be extracted.

27 Relation to Existing Mathematical Structures

HSMT draws on mathematical structures that have appeared independently in several areas of mathematical physics.

These include:

- graded Hilbert spaces;
- Dirac-type operators;
- spectral geometry;
- nonlocal and pseudodifferential operators;
- heat-kernel methods;
- spectral dimension;
- projection and resolvent methods;
- open quantum systems;
- effective field theory.

The purpose of HSMT is not to replace these established structures but to organize them around a particular geometric hypothesis: physical reality may be represented by a hierarchy of coupled effective manifolds, with ordinary physics recovered, if at all, through projection onto an effective grade.

The relationship to established approaches should therefore be understood as structural rather than as a claim of identity.

28 Conceptual Summary

The present HSMT framework can be summarized by the following chain:

Hierarchical geometry $\rightarrow$ Gaussian grade coupling $\rightarrow$ self-adjoint reference operator $\rightarrow$ isolation of a distinguished grade $\rightarrow$ Feshbach/Nakajima–Zwanzig projection $\rightarrow$ effective Weyl dynamics $\rightarrow$ scale-dependent spectral order $\rightarrow$ dimensional flow $\rightarrow$ reduced temporal dynamics $\rightarrow$ gauge and matter extensions.	(112)
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The first several stages can already be formulated mathematically.

The later stages remain the research program.

This distinction is essential to the current maturity of the theory.

29 Conclusion

Hierarchical Shell-Manifold Theory is presently a proposed framework for organizing physical structure through a discrete hierarchy of coupled geometric sectors.

Its fundamental object is the graded structure

$$\mathcal{M} = \bigsqcup_{N \in \mathbb{Z}} \mathcal{M}_N, \quad (113)$$

together with localized inter-grade coupling.

The Gaussian overlap provides a concrete model of residual interaction between neighboring grades. The associative reference realization places the construction on a well-defined Hilbert space and permits explicit operator analysis. Projection onto a distinguished grade produces an exact effective resolvent containing the influence of the eliminated hierarchy through a self-energy term. In an appropriate isolation regime, the projected sector admits an effective Weyl description.

A scale-dependent spatial symbol provides a controlled mathematical mechanism for spectral dimensional flow. The relation

$$d_s = 3/\gamma$$

is established for homogeneous spatial symbols, while a running $\gamma(k)$ provides a route toward an infrared three-dimensional and ultraviolet two-dimensional spatial spectral structure. The crossover scale k_* remains an independent input and must eventually be derived.

The framework also distinguishes effective temporal evolution from the thermodynamic arrow of time. Projection can generate effective non-stationarity, while irreversible dynamics require additional reduced-system assumptions. Thus an effective time parameter and a temporal arrow

are separate problems.

The present formulation does not claim to have derived the Standard Model, gravity, cosmology, or all observed physical parameters. Gauge groups, matter content, flavour structure, and gravitational dynamics remain downstream targets. This is not a limitation of the framework's purpose; it defines the current boundary of the theory.

The central open problem is therefore no longer whether HSMT can be described as a collection of conceptual ideas. It can now be formulated as a mathematical architecture.

The next question is more demanding:

Can the remaining physical structures be derived from the HSMT core without independent postulates? (114)
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If that program succeeds, HSMT would move from a framework for a possible physical theory toward a physical theory itself. If it fails at one of the explicitly identified stages, the framework provides concrete locations at which its assumptions must be revised.

The present paper therefore defines HSMT not as a completed theory of nature, but as a structured theoretical program whose central mathematical mechanisms are explicit enough to be analyzed, extended, and potentially falsified.

A Minimal Reference Operator

For convenience, a compact representative reference operator is collected here.

Let

$$\mathcal{H} = L^2(\mathbb{R}^3, \mathbb{C}^2) \otimes \ell^2(\mathbb{Z}). \quad (115)$$

Define

$$D_{\text{sp}} = -i\boldsymbol{\sigma} \cdot \nabla, \quad (116)$$

and

$$W_N = \kappa \tanh[\alpha(N - 4)]. \quad (117)$$

A nearest-neighbor reference Hamiltonian is

$$H = D_{\text{sp}} + \sum_N W_N \Phi_N + t \sum_N (\Phi_{N+1,N} + \Phi_{N,N+1}). \quad (118)$$

A full Gaussian realization replaces the hopping term by

$$H_{\text{hop}}^{(\sigma)} = t \sum_{N,M} p_{N-M} \Phi_{N,M}, \quad (119)$$

with

$$p_k = \frac{\exp[-k^2/(2\sigma_0^2)]}{\sum_{j \in \mathbb{Z}} \exp[-j^2/(2\sigma_0^2)]}. \quad (120)$$

This operator constitutes the minimal associative laboratory for the next stage of HSMT development.

B Notation and Status Summary

Symbol	Meaning	Current status
$\mathcal{M}_N$	grade- N manifold	foundational definition
N	hierarchical grade	integer label
ρ	continuous hierarchical coordinate	organizing coordinate
σ_0	Gaussian width	candidate/test parameter
W_{NM}	grade-coupling kernel	defined model structure
$\mathcal{H}$	reference Hilbert space	explicit realization
H	reference Hamiltonian	mathematical laboratory
P	projection onto $N = 4$	explicit construction
Q	complementary projection	explicit construction
$\Sigma(z)$	projected self-energy	exact formal result
H_4	effective grade-4 operator	reference-model result
$\gamma(k)$	running spectral order	working ansatz
k_*	dimensional crossover scale	free input
d_s	spatial spectral dimension	derived for specified symbol
t	inter-grade hopping scale	model parameter
v	grade-separation scale	model parameter

C Final Research Statement

The present HSMT program can be stated in one sentence:

HSMT investigates whether ordinary physical structure can be recovered as a controlled projected limit of an integer-graded geometric system whose sectors are coupled through localized hierarchical interactions.

The mathematical task is to determine whether this statement can be closed into a unique, predictive physical theory.

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